Chapter 14

Sound



In our daily life, we hear several sounds, like the sound of birds and animals, the sound of machines and vehicles, the sound of T.V. and telephone, etc. Some sounds are such that we human cannot hear them, but some animals and birds are able to hear it.

There are several questions related to sound like-

1. What is sound? How is it produced?
2. How does sound produced by a source reach our ears.
3. A person standing outside cannot hear clearly the sound of a person sitting inside a closed car, a closed bus or a closed room. But when you knock on the door or window of a car, the sound can be heard on both the sides. Why is it so?

In this chapter, we will try to find answer to all such questions. Also, we will do activities and experiments related to this.

14.1 How is sound produced?

Acitivity-1

1. Take a meter scale.
2. Press one end of the scale on a table for some time, keeping 40 cm of the scale free. (as shown in fig. 1).
3. Now press the free end of the scale and leave it. We see that the scale starts to vibrate and a sound is produced. Now repeat this activity by keeping 30 cm, 20 cm and 10 cm of scale free. What did you experience? Does the length of scale left free affect the sound produced?

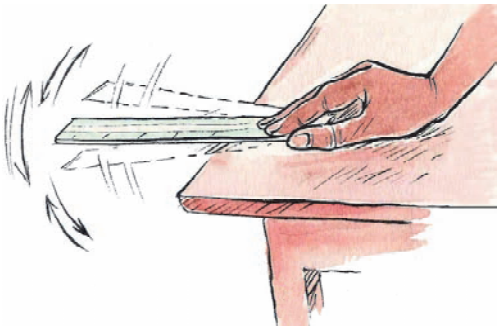


Fig. 1 : Vibration in a metre scale

Activity-2

1. Tie a matallic wire tightly between two hooks (as shown in fig. 2). Note that there should not be any knot in the wire.



Fig. 2 : Metallic wire tied between two hooks

2. Vibrate it using your finger.

Can you hear some sound? If yes, then is this sound different than the sound produced in activity-1.

Activity-3

1. Take a tunning fork.
2. Vibrate this fork by beating it once on a rubber pad.
3. Bring the vibrating tunning fork close to your ear.

Can you hear its sound? Touch one side of the vibrating tunning fork with your finger. What did you experience? Discuss with your friends.

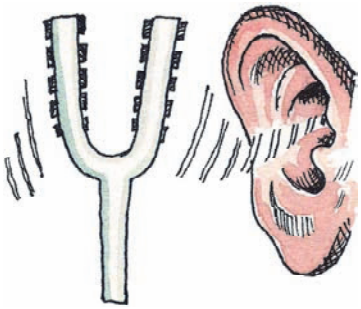


Fig. 3 : Vibrations in a tuning fork

In the above activities you saw that sound is produced in a body due to vibrations. Different types of sounds are produced in different types of objects (materials). Sound depends on the elasticity of a material. Elasticity is that property of a material, due to which it resists any change in its shape or size. That means, when an object changes its shape or size, or tries to change it due to applied force, then the object tries to resist this change and tries to get back to its original shape due to its property of elasticity.

The more elastic a material is, it will try to get back to its original shape quicklier. For example, matallic objects are more elastic as compared to objects made of rubber. Hence it is difficult to pull, bend or change their shape in comparision with rubber. This property of material affects the sound produced from them.

The length of vibrating objects also affects its sound. We cannot listen to the sound of vibration of an object below its minimum required length.

14.2 How does sound reach us?

We saw that sound is produced from vibrating things. When a certain thing vibrates, it exerts a force on the particles of nearby media which are in direct contact with it. So the particles of this medium get displaced from their balanced position or position of rest and start exerting force on the particles of other medium in their vicinity. After displacing the nearby particles, the initial particles get back to their original position. In this manner, energy gets transferred from one particle to another and sound travels forward. This same process continues until sound reaches our ear.

Is sound a form of wave?

Come, lets understand this based on the activity-3 done earlier.

In activity-3 when you vibrate a tunning fork, the arms of tunning fork start to vibrate towards left and right respectively. When they vibrate towards left, they create an area of high pressure by pushing the particles of air surrounding it. This area is called compression (Fig. 4). Now when they again vibrate backwards and return they create an area of low pressure which is called a rarefaction.

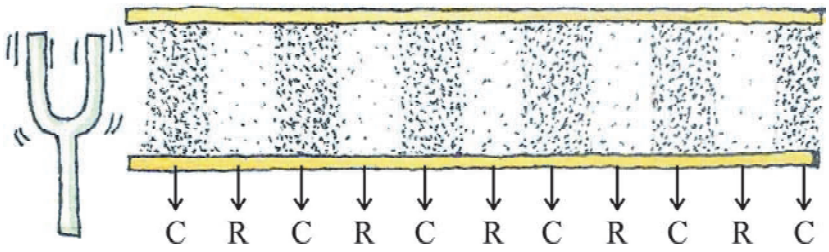


Fig. 4 : L Series of compressions and rarefactions in a vibrating tunning fork.
(C shows compression while R shows rarefaction)

When a tunning fork vibrates continuously, a series of compression and rarefaction is formed in air. These compressions and rarefactions form a wave which transmits through a medium.

As energy is transferred from one particle to another during the transmition of sound therefore, sound is a type of mechanical wave.

Does sound transmission in solids, liquids and gases occur similarly?

Activity-4

1. Put one ear near the top surface of one of the ends of a wooden bench.



Fig. 5 : hearing sound through a solid medium

2. Ask your friend to knock on the other end of the bench.

   Can you hear the sound?

   In this case sound reaches your ears through which medium?

3. Now raise your ear away from the bench, and ask your friend to knock on the bench once again.

   In this situation, sound reached you through which medium?

   What was the difference between the sounds produced in both the situations? Discuss with your friends.

Discuss

1. The sound of a motor boat can be heard louder under the water as compared to above the surface of water.

   Transmission of sound depends upon the density of the particles of the medium. If the density is more, the pressure developed in the medium due to vibration will be more and so the transmission of sound will be quicklier. The particles in solids and liquids are closer as compared to gases. Their density is greater than gases, therefore the displacement of particles due to vibration is quicklier and energy wastage during energy transmission is lesser. For this reason sound travels faster in solids and liquids as compared to gases. In a similar way, transmission of sound in solids is faster as compared to liquids and energy loss is lower, hence we are able to hear the sound clearly.

   In terms of elasticity, such substances in which force of attraction between particles is more, are more elastic as it is difficult to bring any change in the positions of such particles. On creating any disturbance in such particles, they quickly try to come back to their original positions and those particles which quickly come back to their original positions are again ready to move vibrate faster. Such mediums which have higher elasticity (eg. steel) have sound transmission faster as compared to mediums with lower elasticity (eg.- rubber). For this reason, the sound transmission in solids is fastest and in gases it is slowest.

Discuss

1. What effect will be produced on sound if we move upwards above the surface of earth.
2. Can you hear sound in space? Give reasons.
3. On a rainy day, we hear the sound of thunder after we see the lightening, why is it so?

Sound and Medium

Until now we studied that transmission of sound depends upon the density of different mediums. Think what will happen if there is no medium. Will the sound reach you? The following experiment was done to test this.

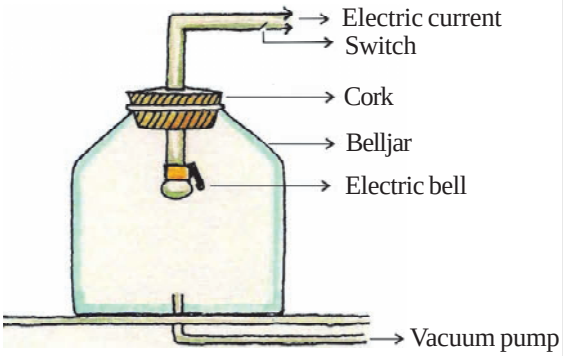


Fig. 6 : Experiment of Belljar

Experiment

An electric bell and a glass belljar was used, as shown in fig. 6. The bell was hung inside the belljar and it was connected to a vacuum pump. On pressing the bell switch, its sound was heard. When the air inside the jar was slowly pumped out using the vaccum pump the sound got lower, even though the same amount of electric current was passed through it. When the complete air inside the bell jar was pumped out, no sound could be heard.

The conclusion made through this experiment was that sound can not travel in vacuum and hence we always need some medium to hear sound.

Sound waves do not travel in vacuum because there are no particles present in vaccum for compression and rarefaction.

Think

If there is no atmosphere on the moon, will astronauts be able to speak to each other.

14.3 Types of waves (on the basis of direction of vibration)

We know that when transmission of waves occur, there is no transfer of particles of medium. They only vibrate with respect to their mean position. On the basis of the direction of vibration of particles with respect to the direction of wave propagation, waves can be of two types-

1. longitudinal waves 2. transverse waves

Activity-5

(a) Formation of longitudinal waves.

1. Take a slinky (plastic spring toy).
2. Hold both the ends of slinky and pull and push them to and fro one by one.
3. Now mark a point anywhere on the slinkey and repeat the above steps. Notice the mark carefully.

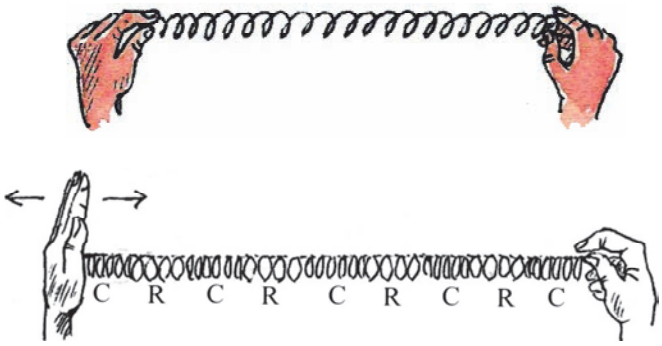


Fig. 7 : formation of longitudinal waves in the slinky (here C is compression and R is rarefaction)

In this activity you saw that on pushing and pulling the slinky to and fro, the mark on the slinky also moves in a direction parallel to the direction of propagation of displacement.

Hence, the waves in which the vibration of particles of the medium is in a direction parallel to the direction of wave propagation, are called as longitudinal waves. The transmission of sound waves takes place in a similar way. Hence, sound waves are also longitudinal waves.

Activity-6

(b) Formation of transverse waves-

1. Take a slinky (spring plastic toy).
2. By keeping one end of the slinky fixed, move the other end up and down.

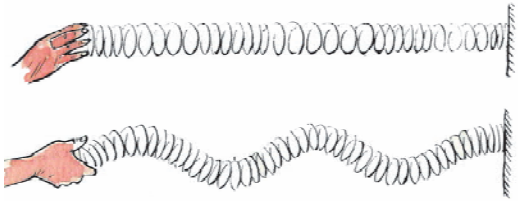


Fig. 8 :

Is the wave formed in this activity different from longitudinal wave?

On moving the slinky up and down, the vibration of particles is vertical to the direction of wave propagation. The wave formed in this activity is called transverse wave.

Hence, the wave in which the particles of the medium vibrate on their mean positions in a direction vertical to the direction of propagation of wave is called transverse wave.

14.4 Characteristics of sound waves

The nature of waves can be defined based on four measures- wavelength, frequency, amplitude and velocity.

Following is a picture showing the sound waves.

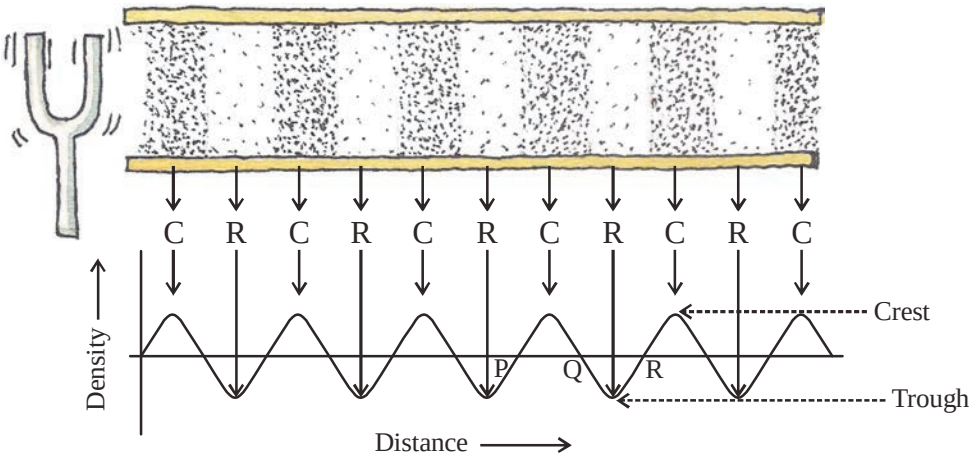


Fig. 9

We know that in a sound wave, compression and rarefaction occur consecutively. The density of air particles in compression is maximum while it is minimum in rarefaction.

This graph is drawn for longitudinal waves between the density of particles in a medium and the distance travelled by a wave. The part PQ in the graph is compression while QR is rarefaction. Similarly the raised part in a transverse wave is called crest and the deep pit is called as through.

Various defining words related to a wave are as following-

1. Wave length

The distance between two consecutive compressions (or crests) or two consecutive rarefactions (or throughs) is called wave length. Its SI unit is metre. It is denoted using λ (Lambda).

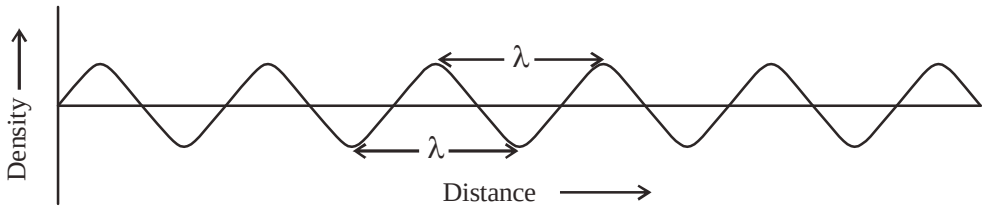


Fig. 10 Wavelenth in density - distance graph

2. Amplitude

When a sound wave travels through air, the particles of air vibrate forming areas of compression and rarefactions. As a result of this, the density of air in an area increases beyond normal and reaches its maximum or sometimes it reduces below the normal and becomes minimum. Therefore, the maximum or minimum densities of particles at the mean position of the medium is called amplitude. It is denoted using the letter A. The loudness or softness of sound mostly depends on the amplitidue of the sound wave.

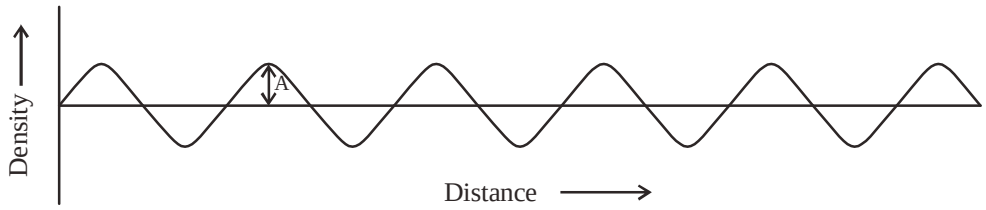


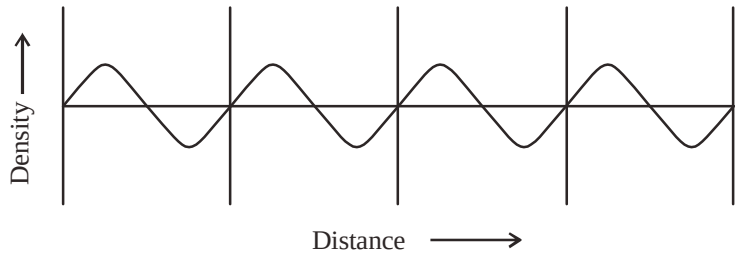
Fig. 11 Amplitude in density - distance graph

3. Time period

The time taken by two consecutive compressions or rarefactions to pass through a point is called time period of the wave. In other words, the time taken for the complete vibration of a density of a medium is called the time period of the sound wave. It is denoted using the letter T. Its SI unit is second.

4. Frequency

We know that when sound travels through a medium, the density of medium changes between the maximum and minimum values. One vibration is completed when the density moves from maximum value to minimum value and then reaches back to maximum value.



Number of vibrations in 1 second = 4

Fig. 12

The total number of such vibrations in a unit time are called as frequency of the sound wave.

If we count the number of compressions and rarefactions passing through the wave in a unit time, then we can find out the frequency of the wave. It is generally denoted by the greek letter υ. Its SI unit is hertz (Hz). The sharpness (pitch) of sound depends upon its frequency.

$$T = \frac{1}{\upsilon} \quad \text{or} \quad \upsilon = \frac{1}{T}$$

Example 1 : Find the time period of a wave of 500 Hz.

Solution : According to the question $\upsilon = 500$ Hz

As we know, time period $T = 1/\upsilon$

$= 1/500$

$= 0.002$ seconds

5. Speed of sound waves

The distance covered by one compression or one rarefaction in a unit time is called the speed of waves.

We know that speed = distance/time

The distance travelled by a wave to complete one vibration is equal to its wavelength and time is equal to time interval.

Hence, $v = \lambda/T$

$\because \upsilon = 1/T$

So, speed $= \upsilon \lambda$

That means speed = frequency × wavelength

Remember that the speed of sound waves only depends upon the nature of medium and does not depend on its wavelength or frequency.

Example 2 : A sound forms 40,000 compressions and 40,000 rarefactions per second while passing through a gaseous medium. If the first compression is formed at a distance of 1 cm from the source, then find the velocity of the wave.

Solution : We know that frequency is equal to the number of compressions and rarefactions in one second.

Hence, frequency υ = 40000 Hz

Wavelength λ = istance between two compressions or two rarefactions

= 1 cm - 1/100 m.

as velocity of waves v $= \upsilon \lambda$

= 40000 x1/100

= 400 m/s

14.5 Hearing range

On the basis of hearing range, sound can be divided into following 3 categories.

1. audible range, 2. infrasound range, 3. ultrasound range

Audible sound- We cannot hear sounds of all frequencies. The hearing range of sound in human is 20 Hz to 20 KHz. We can only hear sound having frequencies within this range. Children below 5 years of age and dogs can hear upto 25 Khz.

Infrasound- The sound having frequency below 20 Hz is called infrasound. For example, the sound produced inside the earth's core at the time of earthquake, chemical and nuclear reaction sounds, etc. If we were able to hear at infrasound range, we could hear the vibrations of a pendulum just like we can hear the vibrations of flapping of wings by a bee hovering near our ear.

Ultrasound- The sounds above 20 KHz are called as ultrasounds. Such sounds are produced by dolphins, bats etc. The hearing organs of insects and moths are very sensitive. They can hear the sound of high frequency produced by bats and thus can protect themselves from bats around them and avoid becoming a prey.

14.6 Applicationis of Ultrasound

Ultrasounds are waves of high frequency. These waves can travel in a straight path even in the presence of obstacles. Ultrasounds are widely used in industries and medical fields.

Ultrasound is generally used to clean parts located in hard to reach places, for example, odd shaped parts, electronic components etc. Objects to be cleaned are placed in a cleaning solution and ultrasonic waves are transmitted into the solution. Due to the high frequency, the particles of dust, greese and dirt get detached and drop out. The object thus gets thoroughly cleaned.

Ultrasounds can be used to detect cracks and flaws in metal blocks. Ultrasounds are also used in the technique of 'echocardiography'. In this technique ultrasonic waves are made to reflect from various parts of the heart thereby forming an image of the heart.

Ultrasonography

Ultrasound scanner is an instrument which uses ultrasonic waves for getting images of internal organs of the human body. A doctor can get an image of the patient's organs such as the liver, gall bladder, uterus, kidney etc. It helps the doctor to detect abnormalities, such as stones in the gall bladder and kidney or tumors in different organs. In this technique the ultrasonic waves travel through tissues of the body and get reflected from a region where there is a change of tissue density. These waves are then converted into electrical signals that are used to generate images of the organ. These images are then displayed on a monitor or printed on a film. This technique is called 'ultrasonography'. Ultrasonography is also used for examination of the foetus during pregnancy to detect defects and growth abnormalities. Ultrasound is also used to break small 'stones' formed in the kidneys into smaller parts.

SONAR (Sound navigation and Ranging)

Sonar is a device that uses ultrasonic waves to measure the distance, direction and speed of under-water objects. Sonar consists of a transmitter and a detector installed on a boat or ship (as shown in fig. 13). The transmitter produces and transmits ultrasonic waves. These waves travel through water and after striking the object on the sea-bed, get reflected back and are sensed by the detector. The detector converts the ultrasonic waves into electrical signals. By finding the speed of sound in water and the time interval between transmission and reception of ultrasound we can find the distance of the object. Let the time interval between transmission and reception of ultrasound signal be t and the speed of sound through water be v. The total distance 2d travelled by the ultrasound is then $2d = v \times t$

Time = distance/velocity

$$t = \frac{2d}{v}$$

Hence, depth $d = \frac{vt}{2}$

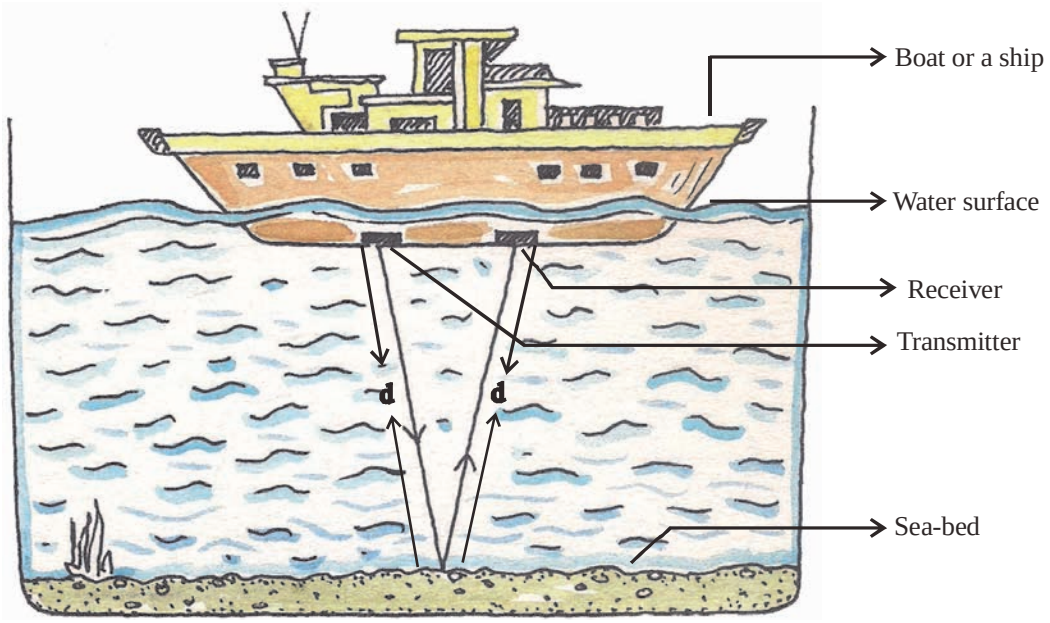


Fig. 13 : SONAR

Example 3 : An investigating team transmits an ultrasound signal to the sea bed. The signal is received back in 6 sec. If the velocity of sound in sea water is 1500 m/s then find out the depth of sea?

Solution : Suppose depth of sea is 'd' m.

So, the distance travelled by signal = 2 d

and time taken = 6 s

speed of sound = 1500 m/s

as Speed $= \frac{\text{Distance}}{\text{Time}}$

So, distance = speed × time

$2d = 1500 \times 6$

$d = (1500 \times 6) / 2$

Hence, depth of sea d = 4500 m = 4500/1000 km = 4.5 km



What we have learnt

- The vibration of a source produces sound.
- Sound travels in a medium in the form of longitudinal waves as successive compressions and rarefactions.
- In sound propagation, only the sound energy travels forward, not the particles of medium.
- Sound cannot travel in vacuum.
- The distance between two consecutive compressions or two consecutive rarefactions is called the wavelength.
- The time taken by the wave for one complete oscillation of the density of the medium is called the time period, T.
- The number of complete oscillations per unit time is called the frequency (υ), $\upsilon = \frac{1}{T}$
- Speed of sound = frequency × wavelength ($v = \upsilon \lambda$)
- The audible range of hearing for human beings is in the frequency range of 20 Hz-20 KHz.
- Sound waves with frequencies below audible range are called 'infrasonic' and above audible range are called 'ultrasonic'. Ultrasound has many medical and industrial applications.
- SONAR technique is used to detect things hidden under the surface of sea.

Keywords

disturbance, wave, wavelength, hearing range, amplitude, freequency, time period, compression, rarefaction

Exercise

1. Choose the correct option-

   (i) When sound waves propagate through a medium, then from one position to another-

   (a) Particles of the medium travel

   (b) Transfer of energy from one particle to another takes place.

   (c) Transformation of energy takes place.

   (d) None of these.

(ii) Sound below the frequency of 20 Hz is called-

(a) amplitude (b) inaudible/infrasound

(c) ultrasound (d) none of these

(iii) The distance between consecutive compressions or rarefactions is called-

(a) amplitude (b) frequency

(c) speed of wave (d) wavelength

(iv) Establish a relation between velocity of transferred wave (v), its frequency and wavelength.

(a) $v = \frac{\upsilon}{\lambda}$ (b) $v = \upsilon\lambda$

(c) $v = \frac{\lambda}{\upsilon}$ (d) none of these

(v) Sound waves do not travel in-

(a) solids (b) liquids

(c) gases (d) vacuum

(vi) If the time period of a vibrating object is 0.05 s, the frequency of waves produced will be-

(a) 5 Hz (b) 20 Hz

(c) 200 Hz (d) 2 Hz

2. Fill the following blanks-

(i) The hearing range of sound in humans is .....

(ii) SI unit of wavelength is .....

(iii) Distance between consecutive compression or consecutive rarefactions is called.....

(iv) Velocity of sound depends on .....

3. What happens to the velocity of sound in a medium if the sound wave frequency is doubled?

4. The frequency of sound produced by source A is twice to the frequency of sound produced by source B. Compare the wavelengths of both the sounds.

5. Which characteristic of sound helps you to identify your friends by his voice while sitting in another room?

6. Sound is produced by colliding two metals, first in air, then in water. Say which medium will produce louder sound when both metals are collided from same distance.

7. What is sound and how is it produced?
8. Explain compressions and rarefaction with help of diagrams.
9. Explain with help of an activity that sound propagation requires a medium.
10. Explain that sound waves are longitudinal waves.
11. Write notes on (i) ECG (ii) ultrasonography (iii) Sonar
12. Explain with diagrams velocity, frequency and wavelength of waves. Also establish a relation between them.
13. A sound wave travels at a speed of 3329 m/s. If its wavelength is 1.5 cm, what is the frequency of this wave? Will it be audible?
14. The speed of sound in air at a certain temperature is 340 m/s and its wavelength is 0.017 m. If the same sound source is dipped in water what effect will be produced on its wavelength? If the speed of sound in water is 1480 m/s, calculate.
15. What steps will you take to reduce the pitch of the sound produced by a 'sitar'?
16. A longitudinal wave whose wavelength is 1 cm, propagates in air at a velocity of 330 m/s. Calculate the frequency of wave. Can a normal human being hear this sound wave?